C) …Binary search is a better option. We start from the middle of the sequence, check if our value (75) is equal to the element in the middle (32), and since it is not, we move right of 32, in the middle of the sub-sequence between 40 and 93. We keep doing so, until either we find a perfect match (which would also be the closest entry) or we end up with a sub-sequence that has just 2 elements. If that's the case, we compare the two elements to our value, and see that 80 is closer than 62: so Happy Harbour is our winner.

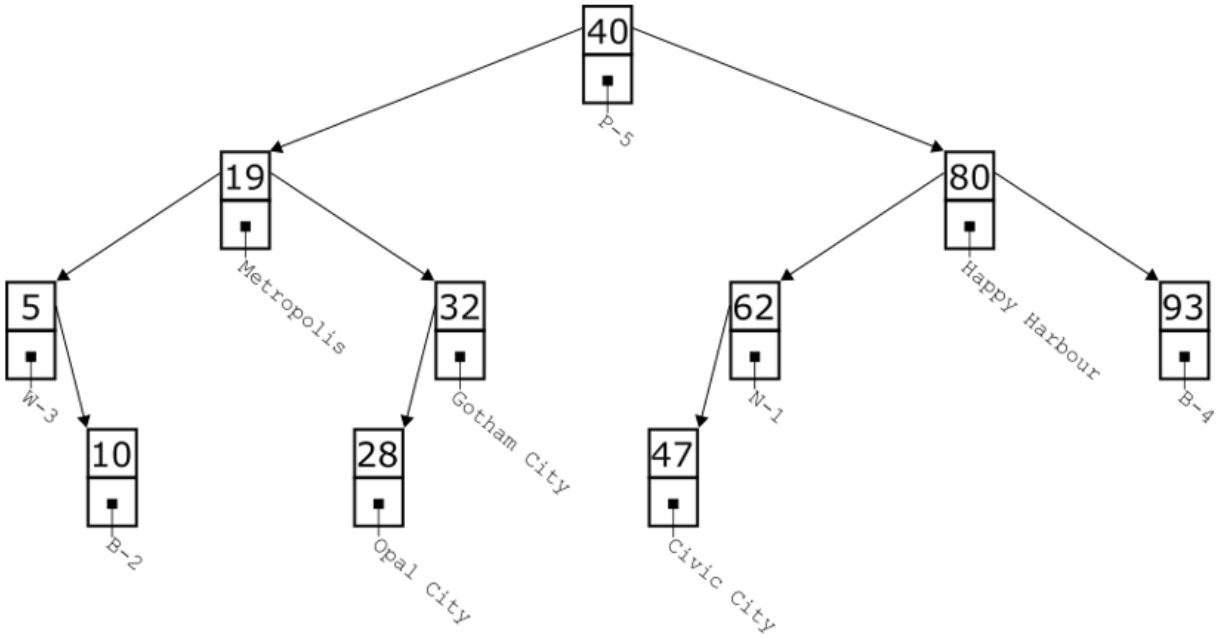


Figure 8.5 A balanced binary search tree containing cities and warehouses in our example. Note that this is only one of the possible balanced search trees for these values: for this set of values, there are at least 32 valid binary search trees that are also balanced. As an exercise, you can try to enumerate all of them (hint: which internal nodes can be rotated[1] without changing the tree height?)